Improving Efficiency and Generalisability of Motion Predictions With Deep Multi-Agent Learning and Multi-Head Attention

Djamel Eddine Benrachou, Sebastien Glaser, Mohammed Elhenawy, and Andry Rakotonirainy

***Abstract*— Automated Vehicles (AVs) have been receiving increasing attention as a potential highly mechanised, intelligent, self-regulating futuristic mode of transport. AVs are predicted to address limitations and human factors associated with traditional modes of transportation. Beyond the typical operations of AVs which can perform rudimentary tasks, the intelligent embedded program fit in to process challenging scenarios and deep multi-dimensional/ agent intents and interaction of the roadway is the grey area yet to be explored to design an exclusive encoding of social functionality and operation in order to address human factors causing road crashes. The aim of this study is to design a data-driven prediction framework for AVs that utilises multiple inputs to prove a multimodal, probabilistic estimate of the future intentions and trajectories of surrounding vehicles in freeway operation. Our proposed framework is a deep multi-agent learning-based system designed to effectively capture social interactions between vehicles without relying on map information. Our approach excels in capturing the high-level behaviours of multiple vehicles and generating a multi-modal trajectory forecast. It employs a multi-headed neural architecture to learn from social interactions between vehicle pairs and generates diverse trajectories proportional to predicted target intents, thus enabling feature fusion. Additionally, a multi-head self-attention mechanism is incorporated for prediction refinement. We achieved a good prediction performance with a lower prediction error in real traffic data at highways. Evaluation of the proposed framework using the NGSIM (US-101 and I-80) and HighD datasets shows satisfactory prediction performance for long-term trajectory prediction of multiple surrounding vehicles. Additionally, the proposed framework has higher prediction accuracy and generalisability than state-of-the-art approaches.**

***Index Terms*— Trajectory prediction, social interaction, driving intention, multi-agent learning, convolutional neural networks, recurrent neural networks, multi-head attention network.**

I. Introduction

AUTOMATED Vehicles (AVs) are imperative and a well-established widespread technology among research, manufacturing firms, engineers, and policymakers. Standardisation and regularisation of the AV engineering technological criterion are still a distant target among the AV community. The specificity of the product design with a multitude of metrics on multiple disciplines to optimise the better-fit prototype is yet to be reached. To recapitulate, a few solutions AVs potentially address are [1], [2], [3]: minimise road crashes, and hazardous emissions, improve fuel economy, increase lane capacities, reduction in travel times and consumer savings and other future technological benefits [4]. AVs require precise equipment for reading, training, and resolving scenarios in order to enhance safety, sustainability, and achieve a futuristic standpoint. Extensive research is conducted to determine the optimal choice of automation capabilities in accordance with the SAE standards.[1] AVs can be seen as intelligent robots with integrated cognitive capabilities equipped with a wide range of high-definition sensors and software components. These technologies work together to create an immersive and realistic representation of driving scenes, enabling a better understanding of the vehicle's surroundings. This, in turn, allows the AV to infer and comprehend high-level behaviours, such as intents and interactions of other roadway participants. Contextually, AVs function to make better intelligent decisions correlating to human decision-making to build a highly intelligent cognisance machine, which will study the interaction between other roadway participants (or agents) in predicting their trajectories to self-iterate the integrated algorithm. The constant dynamic interaction of the other agents influencing the resultant trajectories of the AVs is a highly complex and computational task [6]. This becomes more complex with the increase in prediction targets, especially in high-density traffic flows [7]. In addition, the observation of traffic participants is usually imperfect/impaired and highly noisy due to spatial occlusions [8], sensor impairments, and uncertainties [9], [10], [11]. The motion prediction task involves forecasting the motion of multiple agents, such as vehicles, in the road scene based on their historical observations and considering their spatio-temporal state changes. However, multi-agent motion prediction is a challenging task due to three main factors: (1) *the intervehicular dependence* [12]; (2) *the predictive model uncertainty* [11], [13], [14], [15]; and (3) *the lack of generalisability* [16], [17]. Therefore, they

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[1]The Society of automotive engineers (SAE) has collaborated with the U.S. Department of Transportation (DoT) to develop a classification system for automated driving systems (ADS) based on their ability to assist the human driver. The classification system consists of six levels [5], ranging from 0 (no automation) to 5 (fully automated classification system.)

may fall short in adapting the trajectory time frames to be informative manoeuvring behaviour rather than a structured output proposition. In the context of the current scope, the research team has previously made a significant contribution to the field with the following study [17] and recommends the reader to refer for coherence and continuum.

A. Paper Organisation

The structure of this paper is as follows. Section II reviews related literature and existing research on vehicle trajectory prediction. Section III describes the problem addressed in this study and presents our methodology in detail. Section IV utilises publicly available datasets for training and testing the proposed motion prediction approach, conducting experiments to validate its effectiveness. Section V presents the obtained results, including comparisons, ablation experiments, generalisability performance. Finally, in Section VI, we offer some concluding remarks and highlight potential research gaps, new insights, and future direction.

II. Related Research

Trajectory prediction is challenging due to the high uncertainty arising from unknown driving intents and on-road behaviours of agents. To effectively address this, popular strategies have been adopted to construct multi-agent learning systems capable of adapting to diverse situations using varied techniques. Existing frameworks have given special attention to incorporating agents' spatio-temporal relationships via map encoding and employing social attention mechanisms. Additionally, the adoption of node-centric modelling approaches [18], [19] has gained widespread importance for representing high-level social interactions [15], [20]. Furthermore, utilising latent variables has emerged as a prominent approach to tackle inherent multimodality effectively. Techniques such as generative models, adversarial learning [21], and transformers [22] have garnered popularity in enhancing prediction realism. We refer the reader to the comprehensive survey for an in-depth examination of motion prediction techniques and previous research in the field [17]. In this section, we will delve into the intricate details of these approaches and discuss their merits and defects. The principal focus of literature is categorised into two components [23] (A) *Map encoding* and (B) *Social Attention Modelling*. Map encoding is further subdivided to rasterized encoding and vectorised encoding methods.

A. Map Encoding

1) **Rasterization-based approaches** rely on grid-based operations to model social interaction between agents. Initially, these approaches utilise handcrafted engineering features to model interaction similar to traditional models, such as the social force model [24]. Table I presents the literature alongside the pros and cons. A wide variety of mechanisms was adopted with varied approaches. Researchers have sought to improve existing methods by comparing them with alternative techniques and exploring multiple combinations to determine the most suitable approach for motion prediction using rasterized encoding. However, rasterized encoding can introduce certain limitations in effectively representing varying numbers of input vehicles without imposing a specific ordering. This constraint can hinder the model's ability to maintain consistent prediction performance across diverse scenarios and over long prediction horizons. In addition, it may result in limited spatial precision due to the narrow grid size and alignment with grid cells. Furthermore, a lack of generalisability across different driving conditions is observed which requires attention. More advanced approaches rely on the generation of a top-down Bird's-Eye View (BEV) image representation, providing flexibility in modelling social interactions, temporal vehicle information, and map details. Contextual information can be encoded in separate image channels, typically utilising popular CNN backbones [25], followed by a decoder to extract intermediate-level features and generate final predictions. However, this flexibility comes at a cost. Generating a BEV image requires rasterization, leading to an unavoidable loss of information [26], [27]. Moreover, if the dimensions of the rasterized image exceed a certain size, there is a higher likelihood of significant information loss. Thus, choosing an appropriate grid map size is crucial to avoid overlooking essential interactions and prevent prediction failures when detecting surrounding vehicles (SVs)[2] that aggressively approach the target vehicle (TV) but remain outside the boundaries of the grid map [28]. In an effort to improve prediction performance, [29] slightly modifies the grid map image to prioritise the most influential interacting agents. Other recent approaches employ multi-agent tensor fusion, aligning individual agent features with a BEV image of the driving scene [30], enabling the modelling of interactions between agents.

2) **Vectorised encoding approaches** do not rely on rasterization to model social interactions. For instance, VectorNet [31], LaneGCN [26], DenseTNT [23] and other alternative approaches [32], [33], are some of the approaches adopted in vectorised encoding approaches and presented in Tables I. These methods have fewer limitations though mainly concentrate on goal prediction and high-level road scene modelling. However, they are prone to various challenges, including computational and memory requirements, a restricted range of achievable goals, inaccurate predictions for non-predefined goals, limited granularity in predictions, and sensitivity to anchor selection.

B. Social Attention Modelling

Different approaches to modelling social interactions vary in their effectiveness in capturing temporal and spatial dimensions of the motion. Alongside RNNs, multi-head attention (MHA) based transformer encoder-decoder models [22] have gained popularity in motion prediction [20], [34], [35], [36], [37] due to their effectiveness in handling time constraints. The MHA-based transformer encoder-decoder mechanisms excel in optimising social attention modelling, thus capturing the most relevant features of agent interactions. The models presented in Table I demonstrate the ability to successfully predict multi-modal trajectories that adhere to the social constraints

[2]For the sake of simplicity, we adopt the annotations "TV(s)" to represent target vehicle(s) and "SV(s)" to represent surrounding vehicle(s) throughout the text.

TABLE I

SUMMARY OF WORKS ON TRAJECTORY PREDICTION METHODS BASED ON *Map Encoding* (*Rasterization-Based Methods* AND *Vectorised Encoding-Based Methods*) AND *social Attention modelling*: CLASS, DATASET (DATA), RELATED WORK (WORKS), PROS AND CONS OF EACH WORK AND SUMMARY OF EACH WORK

Class	*Data*	*Works*	*Summary*	*Pros*	*Cons*
Rasterization-based Methods	ETH [44] and UCY [45]	Alahi et al. [46] 2016	S-LSTM is a social generative model for predicting agent motion. It employs individual RNN modules and a social tensor to capture interactions. The social tensor utilises spatially organised hidden states to generate trajectory predictions.	(1).S-LSTM captures spatio-temporal relationships between agents in the road scene;(2).The social pooling strategy captures the essence of the interaction context between the TV and the SV within a specific neighbourhood.	(1).S-LSTM: computationally ineffective and requires handcrafted features designed to setup the interaction; (2).It struggles to capture global context effectively [21]; (3). Fully connected (FC) layer disrupts spatial arrangement in the social tensor; (4).Treating close and distant cells equally in FC hampers generalisability to diverse spatial settings [6].
	NGSIM I-80 and US-101	Deo et al. [6] 2018	Convolutional Social Pooling LSTM (CS-LSTM) combines CNN and RNN layers for effective spatio-temporal data handling. It applies convolutional social pooling to predict vehicle motion on highways.	(1).CS-LSTM improves pooling with convolutional and max-pooling layers for embedding LSTM states in the social tensor;(2).CS-LSTM models the spatio-temporal dependency between agents.	(1).CS-LSTM focuses on successive local interactions followed by a max-pool layer. However, local information is not always sufficient;(2).The generated context vector is independent of the TV state.
	NGSIM I-80 and US-101	Hasan et al. [29] 2021	Polar-LSTM (P-LSTM) incorporates a pooling using polar representation for trajectories, vehicle orientation, and radial velocity. It utilises a pooling strategy to capture social interactions, departing from relying solely on Euclidean trajectory representation.	(1).P-LSTM: predicts driving intents;(2). It generates multi-modal future trajectory distributions;(3).It improves predictions by incorporating polar coordinates, orientation, and radial motion velocity	(1).P-LSTM achieves lower performance compared to CS-LSTM;(2).P-LSTM's representation of vehicle interactions lacks accuracy in terms of interaction weights [47].
	ETH [44] and UCY [45]	Gupta et al. [21] 2018	Social-GAN (S-GAN) demonstrates a slight improvement over CS-LSTM by introducing a diversity loss, which allows for the exploration of a wider range of plausible future trajectories.	(1).S-GAN predicts multi-modal trajectories for other agents; (2).S-GAN does not rely on hand-crafted features [46].	(1). GANs can encounter model collapse, leading to convergence problems and limited diversity in generated outputs;(2). The generative and discriminative components of GANs can cause learning conflicts and lead to unstable training [48, 49].
Vectorised Encoding-based	Argoverse [50] and In-House [31]	Gao et al. [31] 2020	VectorNet pioneered the use of vectored information for lanes and agents, incorporating an undirected fully-connected GNN with polyline-centric nodes to encode social interactions [51]	(1).VectorNet is a hierarchical heterogeneous graph-based approach that models high-level social interactions among agents;(2).VectorNet utilises HD maps to provide a comprehensive depiction of the road.	(1).VectorNet focuses solely on road lane centerlines without explicitly encoding of the surrounding spatial locations; (2) VectorNet's reliance on HD maps makes it resource-intensive; (3).VectorNet has limited generalisability.
	Argoverse [50] and Waymo Open Motion Dataset (WOMD) [52]	Gu et al. [23] 2021	DenseTNT is an end-to-end, anchor-free approach that focuses on goals for multi-trajectory prediction. It generates a dense set of goal candidates with associated probabilities and uses a goal set predictor for determining final trajectory goals.	(1).DenseTNT generates a multi-modal prediction of the trajectories;(2).DenseTNT can handles multiple interacting agents in the roads;(3).DenseTNT is an anchor-free algorithm	DenseTNT uses an offline mode and an optimisation algorithm to generate multi-future pseudo-labels. These labels enhance the adaptability of DenseTNT to scenario-specific operations in the online mode.
	Argoverse [50]	Liang et al. [26] 2020	LaneGCN uses polyline segments to represent map nodes, and graph convolutional network (GCN) to capture road scene topology.	(1). LaneGCN provides an advanced representation of the road environment;(2).GCNs performs convolutions directly on the graph and encode spatial information from neighbouring vehicles	(1).LaneGCN incurs significant computational overhead;(2).The GCN may encounter challenges in adapting to varying neighbourhood sizes while maintaining the weight-sharing properties of the CNN.
Social Attention Modelling	NGSIM I-80 and US-101	Mercat et al. [20] 2020	SAMMP utilises two MHA to capture social interactions and generate joint forecasts for all agents in a freeway scenario.	(1).SAMMP considers uncertainties;(2). SAMMP encodes social interactions among other agents;(3).SAMMP uses a vehicle-to-vehicle attention interaction layer;(4).SAMMP excels in simultaneously handling varying and arbitrary interacting agents without requiring a specific ordering.	(1). SAMMP is computational complex; (2).SAMMP's effectiveness diminishes when predicting on a longer time horizon, as indicated by an RMSE of 2.81m and 3.98m at 4s and 5s, respectively; (3).It does not consider spatial features of agents.
	Argoverse [50]	Gómez-Huélamo et al. [53] 2022	The Attention-based Generative Model (AGM) is a three-channel algorithm that leverages map context, social interactions, and historical trajectories for prediction.	(1). AGM combines RNNs and HD maps to achieve superior performance and handle behavioural interactions;(2).The algorithm integrates both physical and social information.	(1).GAN struggles to cover the full range of trajectories and cannot sample from a distribution[20];(2).The algorithm is computationally complex;(3).The algorithm is limited to medium range;(4).AGM is deterministic.
	Waymo Open Motion Dataset (WOMD) [52]	Shi et al. [51] 2022	The Motion Transformer (MTR-A) is a goal-centric approach that utilises historical agent features to regressively learn and localise the agent's destination among densely sampled goal candidates.	(1). MTR-A is built upon an augmented MHA that learns the spatio-temporal representation of sequential data; (2). MTR-A learns and retrieves scene and social context with time constraints during prediction;(3). MTR-A is a stochastic multi-modal trajectory predictor.	(1). MTR-A uses multiple stacked MHA layers, resulting in computationally burdensome;(2). MLP disrupts temporal consistency in the learning/prediction phases;(3). The prediction accuracy in the goal-oriented scheme relies on the efficiency of goal forecasting [54]-[55].
	Argoverse [50]	Gilles et al. [56] 2021	HOME utilises a 2D top-view grid map to predict the future final point position. The predicted position is then decoded to generate a complete trajectory.	(1).HOME represents the agent's future location as a PDF;(2).HOME uses a shallow CNN with MHA to enhance social interactions;(3).It incorporates two prediction sampling methods, including direct sampling from the heatmap distribution.	(1).HOME's single-point prediction may limit its ability to capture real-world complexity.;(2). HOME's performance may be limited by map accuracy and granularity.
	NGSIM I-80 and US-101 and HighD	Messaoud et al. [34] 2021	The MHA-based LSTM trajectory prediction scheme is a hybrid approach that combines rasterization and attention-based mechanism to operate on freeways.	(1).MHA-LSTM achieves an RMSE of $3.83m$ on NGSIM at a 5-second prediction horizon; (2).It does not rely on HD maps; (3).It enhances the social max-pooling mechanism by incorporating an MHA to capture non-local vehicle dependencies.	The effectiveness of MHA-LSTM may be constrained by its reliance on specific interactions with relevant interactive agents, which limits its applicability across different settings.
	NGSIM I-80 and US-101	Yuan et al. [57] 2021	AgentFormer is a deep seq-2-seq learning-based predictor that jointly designs the spatio-temporal aspect of the traffic.	(1).AgentFormer accurately predicts by attending to last time-step features; (2).It forecasts latent intents for all agents; (3).It is capable of predicting multi-modal trajectories; (4). It integrates latent intent modelling, agent-aware attention, and semantic maps for unified spatio-temporal trajectory prediction.	(1).AgentFormer is computationally resource-intensive; (2).It may not cover all crucial modes comprehensively; (3).It has limitations in controlling predicted trajectory properties. [58].
	Argoverse [50]	Sadeghian et al. [59] 2019	SoPhie is an interpretable framework for predicting trajectories that incorporates social attention. It uses GAN and MHA layers and leverages historical trajectories and local scene context.	SoPhie jointly designs the information about the physical environment and social interactions. It shows a better learning capability with an enhanced prediction performance.	Minor variations in agent interactions and positions may have a significant impact on prediction performance.

of the environment. However, there are a few limitations to consider:

- They assume that only certain SVs interacting in the vicinity of the TV are considered 'effective' while disregarding others that are labelled as 'non-effective' agents, which may not be suitable for making predictions across various traffic scenarios. This limited criterion for selecting effective agents can compromise the performance and generalisability when applied to different datasets and settings.
- They solely focus on predicting the trajectory of one TV at each time step, requiring significant computational power to predict trajectories for all TVs at different time steps.

C. Current Context & Contributions

For the observation, the methods mentioned in Table I have drawn heavily on multi-agent interaction and social robots paradigm [38], [39] to provide motion prediction with a social consistency. However, these methods count some limitations:

1) Most of the previous studies on vehicle trajectory prediction were based on prior locations, and the dynamics of the vehicle model were often used because it was assumed that these features would not suddenly change in a short period during smooth driving [40]. However, in the context of long-term prediction tasks, this assumption is no longer valid, thereby limiting the prediction performance [41]. Modelling based on driving manoeuvre forecast is one of the proposed methods to enhance long-term accuracy;

2) They are effective in the short term but quickly come across performance degradation over a long prediction horizon of 3 $\sim$ 5s;

3) They may show a lack of generalisability—quite a few studies have tried to address the generalisability limitation by developing a generic framework [42] that can operate on different driving scenarios or diverse, realistic driving datasets.

Therefore, there is a shortage of an efficient prediction framework adaptive to different scenarios of unseen model parameters and situations. In light of the aforementioned identified gaps, this study proposes a corresponding approach, as outlined in the following section.

Our contributions. The proposed approach aligns closely with the research conducted in [20], [34], and [43], emphasising the importance of utilising the MHA-based transformer encoder-decoder and considering social interactions to enhance motion prediction with improved interpretability. Unlike the approaches mentioned above, our framework incorporates the robust mechanisms of the CNN-RNN-based dual-stream structure. This structure integrates spatio-temporal features of multiple SVs and TVs and multi-head modules to capture temporal dependencies of trajectories while preserving spatial relationship patterns between traffic agents. Our approach provides a clear and efficient solution to integrate and conserve temporal and spatial context information. A symmetric CNN-based encoder-decoder is designed to capture the spatial dependencies among target agents across various raster grid resolutions. The skip-connected layers are merged using an element-wise summation. The resulting fused target tensor retains its spatial structure, allowing the extraction of the fused target encoding for each target based on its respective grid location. Our approach introduces a bidirectional LSTM (Bi-LSTM) based decoder, which accesses the final output hidden states of all observed vehicles in both directions for each time step during the decoding process. Additionally, we utilise the MHA-based transformer encoder-decoder on top of the neural mixture to model potential future interactions among vehicles, thereby improving prediction accuracy. To demonstrate the efficiency of the developed algorithm, we implemented it on two widely-used large-scale benchmark datasets. As a result, this study aims to contribute to knowledge by integrating various approaches to deduce the best-fit mechanism for improved predictability.

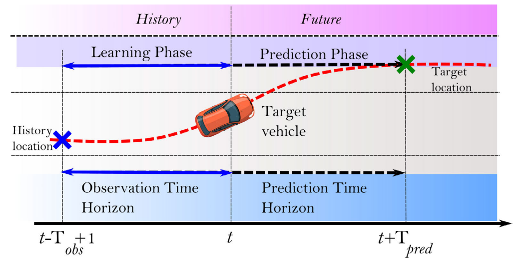

Fig. 1. Trajectory prediction for each predictable agent using the prediction model based on T_{obs} and predicting locations over T_{pred}.

III. Algorithm

This section is organised as follows. Subsection. III-A gives a general definition of trajectory prediction problem of multiple surrounding vehicles. Subsection. III-B describes the pipeline of the proposed approach.

A. Problem Statement

This study approaches the trajectory prediction problem by predicting the forthcoming trajectories of all objects within a given scene, relying on their past trajectories as a basis. The trajectory prediction is framed as a regression problem that aims to predict the plausible trajectory distributions of M interacting agents on the roadway within a fixed time frame (T_{pred}), and relies on the past trajectories of these objects observed during the observation horizon (T_{obs}). We denote the joint states in a temporal range as $\mathbf{Y}_m \doteq \{y_m^{t+1}, y_m^{t+2}, \ldots, y_m^T\}$ and $\mathbf{Y} \doteq \mathbf{Y}_{1:M}^{t:t+T_{pred}}$. The time range considered is $t - T_{obs} + 1$ to t, with T_{obs} being a fixed time horizon at any instant t. These predictions are derived from a feature matrix of previous observations, $\mathbf{X} = \{\mathbf{x}_i^{t-T_{obs}+1}, \mathbf{x}_i^{t-T_{obs}+2}, \ldots, \mathbf{x}_i^t\}$, and $\mathbf{X} \doteq \mathbf{X}_{1:M}^{t-T_{obs}+1:t}$. The aim is to fit a model $\mathcal{H}$ that maps observation to predictions with lower prediction error. Figure 1 shows a simplified depiction of the trajectory prediction problem.

B. Framework Structure

The proposed framework aims to predicting the future motion of a TV by utilising the historical states of the TV and its SVs at different observation time steps. To achieve this objective, the framework leverages machine learning techniques to learn the posterior distribution $P(\mathbf{Y}|\mathbf{X})$ of multiple target trajectories extracted from the area surrounding the TV, referred to as Ψ_{TV}. The conditional elements of this distribution consist of the past trajectories of the SVs around the TV. At any given time step t, the historical trajectory of a vehicle i is represented by the state vector $\mathbf{X} = \{\mathbf{x}_i^{t-Tobs+1}, \mathbf{x}_i^{t-Tobs+2}, \ldots, \mathbf{x}_i^t\}$, where $\mathbf{x}_{i^t} = (x_i^t, y_i^t, .x_i^t, .y_i^t, “x_i^t, “y_i^t)$ represents the longitudinal and lateral position, velocity, and acceleration of vehicle i at time step t. It should be noted that $\mathbf{X}_{\text{TV}}$ refers to the state of the target TV, while $\mathbf{X}_{\text{SV}}$ represents the state of the SV. The waypoint coordinates of all vehicles under consideration are defined within a fixed frame of reference, with the origin positioned at the TVs during different time intervals t_{obs} and t_{pred}. The elements of the state vector are as follows: x_i and y_i represent the waypoint coordinates of vehicle i in the $x-axis$ and $y-axis$, respectively, with respect to the direction of motion on the freeway and the direction perpendicular to it. The framework learns the posterior distribution $P(\mathbf{Y}|\mathbf{X})$ of the TV trajectories, and subsequently generates parameters describing the probability distribution over the predicted locations of the TVs within a finite time horizon, denoted as $\mathbf{Y} = \{\mathbf{y}_i^{t+1}, \mathbf{y}_i^{t+2}, \ldots, \mathbf{y}_i^{t+T_{pred}}\}$. In this representation, the elements x_i and y_i in $\mathbb{R}^2$ depict the past and future waypoint coordinates, respectively. Moreover, T_{obs} and T_{pred} indicate the number of observation and prediction frames, respectively. Finally, $\hat{\mathcal{Y}}^t = \{(x^t, y^t) \in \mathbb{R}^2 | t = 1, 2, \ldots, T_{pred}\}$ represents the predicted waypoint coordinates of visible/observable TVs. The framework assumes that the conditional probability distribution $P(\mathbf{Y}|\mathbf{X})$ follows the bivariate normal distribution [6], [60].

The overall flow of the prediction scheme follows this sequence: 1) ***Encoders***, 2) ***Social Interaction***, 3) ***Manoeuvre-based decoder***, 4) ***Information Fusion***, and 5) ***Decoders***

As shown in Fig. 2, the prediction scheme is composed of five parts. Each part is described in detail as follows:

1) ***Encoders:*** (see Fig. 2 1) In furtherance to achieve realistic predictions about vehicle motions, it is crucial to understand the spatio-temporal relationship and social interactions among vehicles in the road scene. In conducting this task, a series of stepwise procedures is adopted. Firstly, all trajectories are encoded independently to enable the algorithm to learn the sequential locations' temporal properties. Secondly, each trajectory is pre-processed by converting its locations into coordinates relative to the TV. These converted coordinates are then passed through a temporal convolution layer, specifically a 1D convolutional layer, to achieve motion embedding. Regarding the first step, the network creates a temporal embedding layer for temporal encoding. This encoding layer incorporates the temporal information of the TVs from the trajectory sequence into the input trajectory feature. As a result, the temporal encoding incorporates the temporal information of TVs from the trajectory sequence into the input trajectory feature, effectively capturing the time-related dynamics. The activation function LeakyReLU$(\cdot)$ has been chosen for the 1D convolutional layers during the initial encoding of the TV dynamic parameters.

Successively, the trajectory encoder encodes the trajectories of SVs that belong to the neighbourhood of each TV at different t_{obs}. The proposed algorithm transcends the limitations of previous approaches [61], [62], [63], [64]. Unlike these approaches, it is not bound by a fixed number of immediately observable SVs around a TV. Furthermore, it avoids reliance on a predetermined SV structure [65]. This distinct characteristic endows the algorithm with enhanced versatility, making it well-suited for real-world driving scenarios where the number of SVs varies [66]. Therefore, it is more prudent to take into account the variable number of observable vehicles [28]. A spatial grid $\Psi_{\mathbf{TV}}$ is then constructed to capture motion information of all observable SVs around each TV. Here, $\Psi_{\mathbf{TV}} \in \mathbb{R}^{H\times W}$, where H represents the height and W represents the width of the spatial grid. This grid is computed over the neighbouring area of each TV. The use of a spatial grid is appropriate for two reasons: (1) it allows for capturing the context of the derivable road, and (2) it considers all SVs in the neighbourhood of each TV at the respective time frame. In this paper, the predicted targets located within the $\Psi_{\mathbf{TV}}$ grid with dimensions of $60.96 \times 10.57m$. This means that the spatial grid covers a longitudinal distance of $60.96m$ and a lateral distance of $10.57m$, discretised into (25×5) pixels [28]. The target-centric area $\Psi_{\mathbf{SV}}$ of each predicted vehicle is built in the same way as $\Psi_{\mathbf{TV}}$. Each state vector $\mathbf{x}_i^t$ of a vehicle i of the neighbouring area $\Psi_{\mathbf{SV}}$ is embedded using a $FC(\cdot)$ layer to yield an embedding vector e_i^t.

$$e_i^t = FC(\mathbf{x}_i^t; W_{emb})$$

where $FC(\cdot)$ is fully connected layer with a LeakyRely activation, W_{emb} is the learnt the embedding weights. The GRU encoder is fed by the embedding vectors of each vehicle i for different observation time steps $t = 1, 2, \ldots, t_{obs}$. The hidden states h_i^t are computed using the formula:

$$\_, h_i^t = GRU(h_i^{t-1}, e_i^t; W_{encoder})$$

The hidden state vector of the GRU is set as h_i^t computed for the $i-$th SV at time t. $W_{encoder}$ is the weight of the GRU encoder. It is worth noting that TVs trajectories are also encoded with a GRU encoder to compute the hidden state vector $h_{\mathbf{TV}}^t$, with shared encoding weights, i.e., the GRU encoders of SVs and TVs share the weight values $W_{encoder}$.

The target encoder's shortcut structure is joined to that of SVs, which can ease the learning of hidden patterns and enhance the learning procedure. Separate branches complete various tasks after the network extracts features from shared hidden layers.

2) ***Social Interaction:*** (see Fig. 2 2) The spatio-temporal relationships among interacting vehicles in a road scene are captured by incorporating a social pooling strategy and a convolutional social pooling structure into a GRU encoder. While the GRU encoder can effectively capture the temporal structure of a trajectory sequence, it can struggle, however, to handle the complex spatial interactions between other vehicles. The social pooling strategy addresses this limitation by pooling the recurrent states of spatially proximal sequences into a social tensor, represented as a target-centric grid. The convolutional social pooling [67] structure further enhances this approach by applying two layers of 2D-CNNs and max-pooling layers over the social tensor. The CNN-RNN-based dual-stream structure can extract a wide range of spatial patterns from feature vectors while capturing different temporal patterns across time steps. By employing these two strategies alongside the GRU encoder, it becomes possible to effectively model the spatio-temporal relationships among interacting vehicles in the traffic scene. These techniques offer a means to overcome

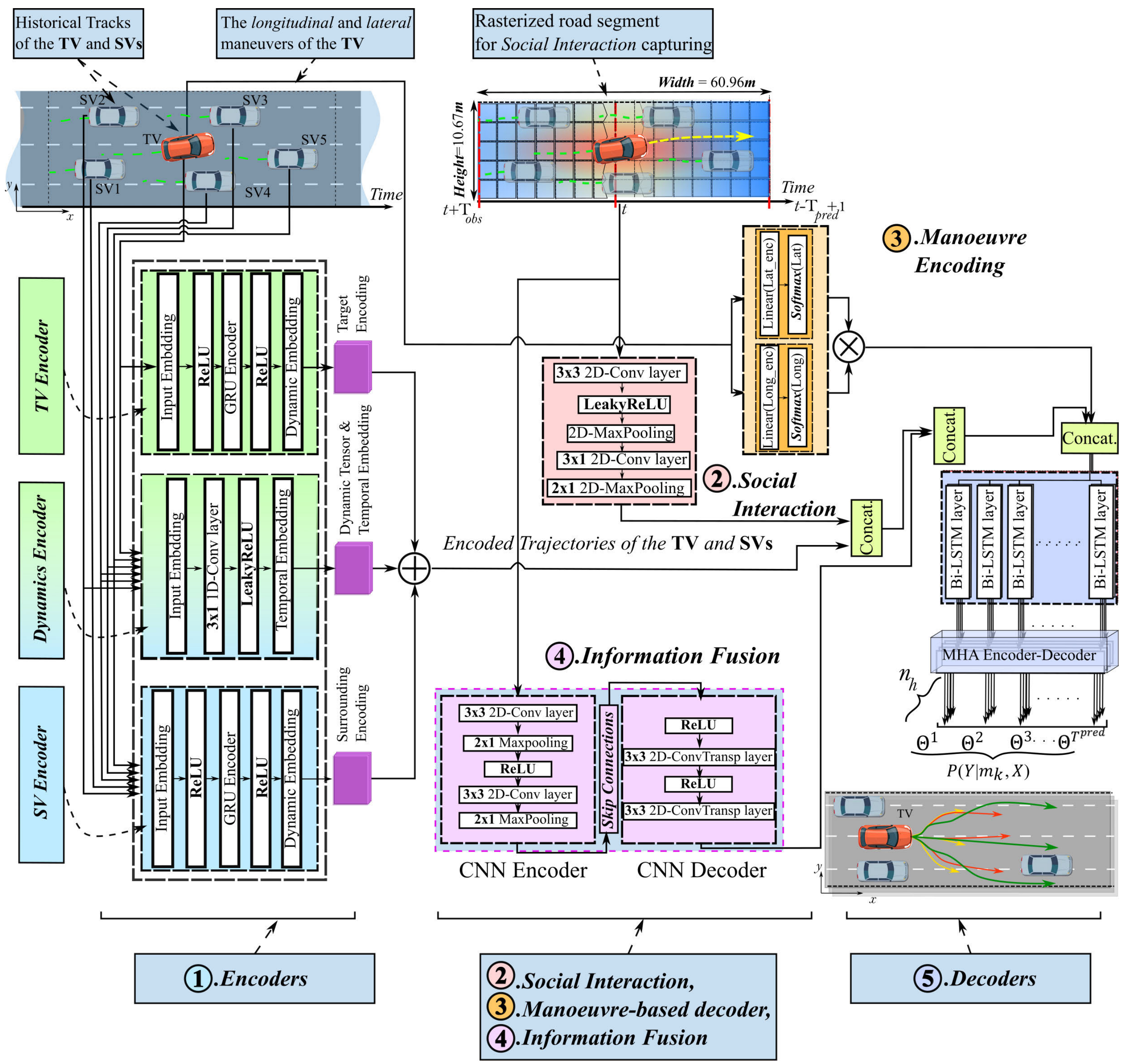

Fig. 2. Overview of the proposed algorithm.

the limitations of conventional GRU encoders, enabling improved modelling of social interactions.

3) ***Manoeuvre-based decoder:*** (see Fig. 2 (3)) In order to address the stochastic and multi-modal nature inherent in driving behaviour, a multi-modal manoeuvre predictor is developed by leveraging the research conducted in [6]. Two predefined sets of manoeuvre categories denoted as $M = \{m_j | j = 1, 2, \ldots, 6\}$, are defined and utilised to model driving intents. These sets encompass lateral intents corresponding to location-based manoeuvres (such as lane-keeping, left lane changes, and right lane changes) and longitudinal intents corresponding to acceleration-based manoeuvres (including normal driving and braking). We refer to the methods outlined in [6] to extract the various driving manoeuvres. The encoded spatial information of TVs is inputted into a pair of $FC(\cdot)$ layers connected to two separate $Softmax(.)$. These layers are responsible for generating probabilities of lateral and longitudinal intents. The $Softmax(.)$ function is applied to normalise the input matrix, ensuring that the resulting weight matrix contains elements bounded between 0 and 1. Additionally, the elements of the weight matrix sum up to 1, providing a valid probability distribution. The multiplication of these values yields the probabilities $P(m_j|\mathbf{X})$ for observed pairwise manoeuvres. To condition the trajectory on the manoeuvres, the encoded convolutional spatial information is concatenated with one-hot vectors representing lateral and longitudinal manoeuvres. The distribution over future trajectories can be hierarchically factorised by first estimating the uncertainty of the intended manoeuvre classes. By incorporating the intended manoeuvres, the future trajectory can then be designed as a

mixture of Gaussian distributions. Our approach structures the intention prediction module akin to a regression task, allowing it to learn and encode probability distributions of manoeuvres instead of relying solely on classification [61]. This mitigates potential biases that may arise from classification errors [17].

4) ***Information Fusion:*** (see Fig. 2 ④) To effectively capture the spatial correlations among future states of the TVs, the proposed approach integrates a target information fusion mechanism [28]. This mechanism utilises a CNN encoder-decoder while preserving the original text structure. By leveraging the CNN's capability to maintain spatial invariance and encode contextual scene features, the network effectively handles spatial grid maps, consolidating motion information and spatial dependencies. The CNN encoder-decoder is responsible for extracting low-order features that prioritise spatial information [28], facilitating the capture of correlations among vehicles. Through the encoding process, target-fusion data is generated, enabling joint prediction within the target area. The symmetric structure of the CNN, complemented by skip-connected layers, efficiently processes target-centric grid maps, merging and fusing tensors to extract a comprehensive fused target encoding that encompasses grid locations, providing a holistic perspective.

5) ***Decoders:*** (see Fig. 2 ⑤) Human drivers are guided by multiple goals, intents, and behaviours, which result in a range of plausible future modes or trajectories. To account for uncertainty, drivers tend to focus on the trajectories of the closest SVs in their target lane. As a result, from each set of SV environments, multiple plausible future states can be derived in a closed form. In the case of $P(\mathbf{Y}|\mathbf{X}, z_t)$, where z_t denotes contextual information at time t and $\mathbf{X}$, there exists the potential for multiple distinct modes, resulting in a mixture of probabilistic distributions encompassing diverse output sequences of $\mathbf{Y}$. However, designing $P(\mathbf{Y}|\mathbf{X}, z_t)$ jointly in a sequential modelling setting can be inefficient and intractable. Typically, a cascade-like structure using RNN and back-propagation through time (BPTT) is applied to model sequential distributions. Nonetheless, this approach often leads to mode-averaging, as mapping information from input $\mathbf{X}$ to output $\mathbf{Y}$ becomes a one-to-many mapping task for handling multi-mode prediction. To handle multimodality, our approach applies a deep Bi-LSTM [68] in conjunction with a MHA mechanism. This approach enables the generation of diverse representations of encoded signals using multiple attention heads. The decoding process consists of two stages: (1).Bi-LSTM Decoder; and (2).Multi-Head Attention-based Transformer Encoder-Decoder.

a) *Bi-LSTM Decoder:* Unlike the traditional method of directly predicting the exact future locations of a vehicle at any given time, which often results in substantial prediction errors as the prediction horizon extends, the current study takes a different approach. It utilises an alternately trained Bi-LSTM decoder that operates in a residual learning way, generating displacement values between predicted locations. By merging the information obtained from observing SVs the resulting encoding encompasses the social context within the previous time domain. The merged social encoding, known as the social context, is then concatenated with the dynamics encoding of the target, represented as $e^t_{\mathbf{TV}}$, to form a target encoding $h^t_{\mathbf{TV}}$. This target encoding aggregates all the accessible information within the $\Psi_{\mathbf{TV}}$. After encoding the past state of each vehicle as h^t_i, a Bi-LSTM is utilised as the first decoding stage. It takes input from the context vector z, which contains information about the interactions between vehicles, and ρ, which contains fusion information and motion encoding of the TV, represented as $h^{t_{obs}}_{\mathbf{TV}}$. The decoding stage can be represented as $h_{decoder} = \text{concat}(h^{t_{obs}}_{\mathbf{TV}}, z, \rho)$. This process generates predicted parameters for the distributions over the estimated future locations of the TV at each time step $t = 1, 2, \ldots, t_{obs}$.

b) *Multi-Head Attention-based Transformer Encoder-Decoder:* In the current stage, we employ a complex interaction interpreter based on the MHA-based transformer encoder-decoder. This approach effectively preserves diverse spatial-temporal relationships encompassed within the Bi-LSTM hidden state. By leveraging MHA, our model refines predictions by extracting relevant interaction features and generates an attention-based representation. The adoption of MHA finds inspiration in the Transformer [22]| with outstanding for its architectural flexibility [69]. Due to the limitation of a single learned attention cue focusing on only one interrelated observable subset of vehicles at each time step t, it may capture only a single facet of the possible spatio-temporal relationships within the convolutional spatial area centred around each TV. However, in many cases, there are multiple aspects or facets that a sequence element needs to attend to, and there are more suitable approaches to handle such specificity than a single learned attention cue. The attention mechanism provides a solution to address a higher level of interaction while effectively leveraging its internal structure as a learning tool. The MHA mechanism simultaneously computes multiple different subsets of queries (Q_l), keys (K_l), and values (V_l) n_h times using n_h attention heads. Each attention head utilises distinct learned linear projections Q_l, K_l, and V_l, where $l \in [1, n_h]$ represents the attention head index. The attention mechanism generates n_h sets of attention features, each showcasing subsets of interrelated SVs alongside the TV. These attention features capture possible spatio-temporal relationships occurring between the TVs. Afterwards, the attention features generated are combined by concatenation and dynamically weighted to extract salient information from subsets of vehicles.

$$\Lambda(Q, K, V) = concat(head_1, \ldots, head_{n_h})W^O$$

where $\Lambda(.)$ with the learnable attention parameters W^Q, W^K and W^V that will be learned in each attention head l. To provide inputs to $\Lambda(.)$, the hidden states of the decoder Bi-LSTM, which are constructed based on the observation time, are utilised as input for the attention mechanism. These hidden states are abstract, high-dimensional representations of the encoded input sequences. When considering all attention heads, the hidden states of the TVs are mapped using Q, K, and V. Dot product attention takes queries Q, keys K, and values V, all of shape $\mathbb{R}^{B\times T\times d}$, where B is the batch size, T is the sequence length (set to $T = 25$), and d is the hidden dimensionality set at $d = 64$. Matrix multiplication QK^T computes dot products between all pairs of elements in Q and K, yielding a $T \times T$ matrix of attention logits. Softmax is applied to this matrix, ensuring proper probability distribution across rows, and the resulting weights are element-wise multiplied with V to compute the weighted mean.

Unlike the approach of using a $FC(\cdot)$ followed by a nonlinear activation function to predict vehicle motion [28], [34], which is limited to learning time series inputs and cannot effectively capture the temporal relationship between previous and current elements in a sequence, the proposed solution in this study addresses this limitation. By utilising an MHA-based transformer encoder-decoder to generate final prediction, long-range spatio-temporal dependencies can be captured, ensuring the retention of temporal consistency in the data. This approach is convenient for long-term motion prediction. The MHA network enhances interactions between the TV and its SVs in the spatial grid map. By selectively attending to subsets of SVs during the computation of the TV's future trajectory, the model can better capture the relevant information and dependencies necessary for accurate prediction.

$$\Theta^t = \Lambda(\text{Bi-LSTM}(h^{t-1}_{decoder}; W_{decoder}))$$

where $\Theta^t i$ is the predicted parameters of the distribution of vehicles' positions at any time step t. $\Lambda(.)$ denotes the Multi-Head Attention network, and $\Lambda(.)$ is the learned weights of the Bi-LSTM decoder. Additionally, $h^{t-1}_{decoder}$ corresponds to the hidden state vector of the decoder at time $t-1$. Our algorithm is trained by minimising the negative log-likelihood loss $\mathcal{L}_{nll}$ of subsequent trajectories while considering all the TVs' ground truth manoeuvre classes.

$$\mathcal{L}_{nll}(\mathbf{Y}_{pred}) = - \sum_{t+T_{obs}+1 \leq t \leq t+T_{pred}} \{log(P_{\Theta^t}(\mathbf{Y}^t|m, \mathbf{X}))\}$$

At the final stage, the generated distribution is represented by a three-dimensional vector field. $\delta y_i^{t+T} \in \mathbb{R}^2$ denotes the displacement vector between neighbouring predicted locations, $\sigma_i^{t+T} \in \mathbb{R}^2$ represents the standard deviation vector, and $\psi_i^{t+T} \in \mathbb{R}$ is the correlation coefficient of the estimated future locations $\hat{y}_i^{t+T}$ over a fixed (future) time horizon $T \in \{1, 2, \ldots, T_{pred}\}$. The distribution over the possible locations at time $t \in \{t - T_{obs} + 1, t - T_{obs} + 2, \ldots, t\}$ can be depicted as a Bivariate Normal distribution.

$$\hat{y}_i^{t+T} \sim \mathcal{N}(\mu_i^{t+T}, \sigma_i^{t+T}, \psi_i^{t+T})$$

where μ is the mean vector, computed by summing up all displacements along the future time frame T from the location at the last time step t of the past trajectory.

$$\mu_i^{t+T} = x_i^t + \sum_{\tau=1}^{T} \delta y_i^{t+\tau}$$

The Gaussian parameters along all future time steps of the TV are denoted as Θ_i, where $\Theta^{t+T} i = (\mu i^{t+T}, \sigma_i^{t+T}, \psi_i^{t+T})$. The posterior probability of all TVs' trajectories can be written as:

$$P(\mathbf{Y}|\mathbf{X}) = \prod_{v_i \in V_{tar}} \sum_{k=1}^{|M|} P_\Theta(Y_i|m_k, \mathbf{X}) P(m_k|\mathbf{X})$$

It yields a Gaussian Mixture distribution (GMM) [25]. The probabilities of different manoeuvres define the mixture weights. Then, the prediction of future manoeuvres (both lateral and longitudinal) will be selected based on the maximum observed likelihood to determine subsequent intents.

IV. Experimental Settings

A. Datasets

We verified our proposed method with two publicly available datasets of naturalistic human driving collected on highways, the NGSIM dataset [70] and the HighD dataset [71]. Provided below are succinct descriptions of the two datasets employed in our study.

- **The NGSIM datasets**. In this work, we utilise both NGSIM I-80 and US-101 datasets [70], which were collected from the highways of Emeryville and Los Angeles, respectively. The datasets were sampled at a frequency of 10Hz (10 frames per second) over a duration of 45 minutes, resulting in more than 8 million trajectories.
- **The HighD dataset**. [71] is a vehicle trajectories dataset released in 2018. The data were recorded from six locations on German highways for aerial perspective using an uncrewed aerial vehicle (UAV). The data encompasses 60 recordings over $400 \sim 420$ meters span, with more than 100000 vehicles contained. In this work, the HighD dataset has been pre-processed to have a similar format to the NGSIM dataset for convenience.

The two datasets are divided into 70% of the training subset, 20% of the validation subset and 10% of the testing subset, following to the methodology outlined by Deo and Trivedi [67]. Each trajectory is segmented into sequences lasting 8 seconds, where the initial 3 seconds serve as the observed track history, while the subsequent 5 seconds constitute the ground truth. To align with the principles presented in [67], we performed a resampling operation, converting the sequence frequency from 10Hz to 5Hz. Specifically, we have downsampled the HighD dataset by using an interpolation process.

For detailed insights into the datasets employed in our experiments, please refer to Table II.

B. Evaluation Metrics

1) **Metrics of Trajectory and Manoeuvre Predictions:** In the evaluation stage, two metrics are used, namely Root of the Mean Squared Error ($\mathcal{E}_{RMSE}$) metric and Negative Log Likelihood ($\mathcal{E}_{NLL}$). These are critical metrics to determine the performance difference.

- **Root Mean Squared Error ($\mathcal{E}_{RMSE}$).** $\mathcal{E}_{RMSE}$ measures the e standard deviation of the residuals (predicted trajectories) which is a measure of how far from the regression line are the data points which are ground truth trajectory data points in this case.

$$\mathcal{E}_{RMSE} = \sqrt{\frac{1}{t} \sum_{t=t+T_{obs}+1}^{T_{obs}+t} (x_T^t - x_{pred}^t)^2 + (y_T^t - y_{pred}^t)^2}$$

- **Negative Log Likelihood ($\mathcal{E}_{NLL}$) or Cross-Entropy:** The $\mathcal{E}_{NLL}$, also known as Cross-Entropy, is used to assess the prediction data distribution and the ground truth data distribution. It can be applied to both the prediction of intention

TABLE II
PARAMETER SETTINGS FOR TRAINING, TEST AND VALIDATION STAGES FOR THE TWO DATASETS NGSIM (I-80 AND 101) AND HIGHD

Dataset	*Frequency*	*Hist. length*	*Fut. length*	*Train Size*	*Validation size*	*Test size*
NGSIM	5Hz	3s (*Seq. length* 16)	5s (*Seq. length* 25)	80457	20627	20626
HighD	5Hz	3s (*Seq. length* 16)	5s (*Seq. length* 25)	59111	7751	15745

and trajectory. The $\mathcal{E}_{NLL}$ is particularly effective in evaluating multimodal predictions. In the case of a multimodal prediction, where there are multiple possible outcomes (e.g., a mixture of a finite number of Gaussian distributions), the $\mathcal{E}_{NLL}$ penalizes the prediction for not covering all the possible modes [17].

$$\mathcal{E}_{NLL}(\phi, \psi) = \mathop{\mathbb{E}}_{x\sim\psi} - log(\psi(x^t_{pred}))$$

where ϕ and ψ represent the trajectory distribution and the ground truth data distribution, respectively.

We use mean values of predicted distributions for subsequent trajectories to compute $\mathcal{E}_{RMSE}$ and $\mathcal{E}_{NLL}$ over a 5-second prediction horizon.

C. Implementation Details

• *Encoder (SV) and Encoder (TV)*: use a GRU encoder with 64 units.

• *Dynamic Encoding.* uses a linear transformation with encoder dimension as input and 32 as output. $ReLU(\cdot)$ activation follows while trajectories are converted to temporal embedding using 1D convolution before the transformation.

• *Motion Social Pooling.* applies a 2D CNN Layer with encoder dimension input, 64 output channels (social depth), and a 3×3 convolution kernel. It includes a max-pooling Layer (3×3 kernel, stride 2) and a second CNN Layer (64 input channels, 16 output channels). LeakyReLU activation is used between the CNN layers.

• *CNN encoder-decoder* (TV). uses a CNN encoder-decoder with a symmetric configuration, two max-pooling and two up-sampling operations. Each CNN layer has a 3×3 filter and a padding and stride of 1 in the first two encoding layers with a $ReLU(\cdot)$ activation for faster training.

• *The Intention Predictor.* fuses information from TV and passes it through $FC(\cdot)$ and $Softmax$ layers to calculate lateral and longitudinal intent probabilities. Multiplying these probabilities gives the manoeuvre class probability $P(m_k|\mathbf{X})$. The trajectory under each class is obtained by concatenating the fused encoding with one-hot intent vectors, and passing it through BiLSTM decoder.

• *BiLSTM decoder.* uses a BiLSTM decoder with 128 units.

• *Multi-Head Attention network.* utilises two heads (h=2) for the HighD data and four heads (h=4) for the NGSIM data. The hidden states' dimension of the Multi-Head Attention is set to 128.

• *BatchNorm.* is applied between the convolution/deconvolution and activation layers, mitigating distribution shift, covariate drift, loss oscillation, weight saturation, and slow learning. This enhances the robustness of parameter initialisation and accelerates convergence to smoother solutions [72].

In our experiments, we set T_{obs} to 3s and T_{pred} to 5s. The method is implemented with PyTorch 3.7.9 [73] on a QUT-provided HPC service, utilising CUDA 11.4, cuDNN 8.2, and the Linux platform for efficient computation. Hyperparameter tuning systematically evaluates parameters to identify the optimal model configuration. To fine-tune our model, we initially train it for five epochs using a learning rate of 10^{-3} and continue training for an additional ten epochs using the Adam optimiser [74].

D. Comparison With Baselines

Our algorithm's performance is evaluated by comparing it to state-of-the-art methods and various control settings, considering vehicle social interactions. All comparative models undergo end-to-end training, except for the HMNet [75], which leverages pre-trained weights available online. It is worth noting that models such as S-LSTM, CS-LSTM, P-LSTM, and S-GAN codes are solely accessible online with compatibility restricted to the NGSIM dataset loader, lacking support for the HighD dataset loader. Consequently, these models have undergone exclusive end-to-end training on the NGSIM dataset without any adjustments made for the HighD dataset. We anticipate addressing this limitation in our forthcoming research to enable a more in-depth evaluation of prediction performance.

• *Manoeuvre-LSTM (M-LSTM)* [46]: an encoder-decoder model designed to encode vehicle trajectories and manoeuvres, which are then passed to the decoder to generate multi-modal trajectory predictions. However, M-LSTM primarily focuses on manoeuvre encoding and does not incorporate information fusion.

• *Social-LSTM (S-LSTM)* [67]: employs a fully connected layer to perform social pooling, resulting in a distribution of future locations that focuses on a single mode.

• *Convolution Social-LSTM (CS-LSTM)* [76]: utilises convolutional layers combined with social pooling and generates a multi-modal prediction based on manoeuvres.

• *Polar-LSTM (P-LSTM)* [29]: uses a pooling strategy with polar representation for trajectories, orientation, and radial velocity to capture social interactions. It predicts vehicles' trajectories in a Bivariate Normal distribution.

• *Social-aware Generative Adversarial Network (S-GAN)* [21]: uses a GAN-based framework that leverages an adversarial loss to produce a variety of trajectories for multiple agents in a spatial-centric manner.

• *Hierarchical Motion Encoder-Decoder Network (HMNet)* [75]: utilises separate LSTM encoders for position, velocity, and acceleration, along with a goal-embedded decoder. It employs hierarchical predictors and a CVAE decoder to learn agents' dynamics separately and predict multiple trajectories.

• *Planning-informed Trajectory Prediction (PiP)* [28]: is a multi-agent learning model that generates joint agent motion predictions conditioned on a planning motion.

• *The Multi-head Attention LSTM (MHA-LSTM)* [34]: utilises an LSTM encoder-decoder with an enhanced social pooling module incorporating a multi-head attention mechanism.

The attention layer captures social interactions prior to the LSTM decoder, which generates the final prediction using an $FC(\cdot)$ layer.

- ***Ours***: The proposed model in this work.

It is worth mentioning that the MHA-LSTM model [34] has been re-implemented from scratch without optimising the internal neural structure.The model includes 64 LSTM units for encoding, a 128 LSTM decoder, and a 32-dimensional embedding space. The projected vectors undergo a dot-product attention operation with the Adam optimiser [74] applied, utilising a batch size of 64. The re-implementation is done in PyTorch, and the special area, centred on the target vehicle position, is expanded to $(90.96 \times 10.6)m$ to ensure that a performance comparison between systems is fair.

V. Evaluation

This section evaluates the proposed approach in three aspects: (A). Prediction efficiency; (B). Ablation experiments; (C). Generalisability test; and (D). Computation Time.

A. Prediction Efficiency

This subsection presents and discusses the effects of using multiple attention heads for optimal prediction efficiency and the obtained results of the proposed approach on the NGSIM and HighD datasets.

1) To attain a comprehensive understanding of the efficacy of the proposed model, which integrates a multi-headed transformer encoder-decoder framework, across the NGSIM and HighD datasets, we systematically calibrated the number of attention heads during training and evaluation. This meticulous approach allowed to pinpoint the optimal configuration: a judicious allocation of attention heads that not only maximised predictive accuracy but also achieved a balance between prediction time and memory utilisation [77]. The validation of the methods was conducted by assessing $\mathcal{E}_{RMSE}$ values over a 5-second prediction horizon using the proposed algorithm on the designated datasets with 2, 4, and 8 attention heads (See Table III). The analysis suggests that configuring the number of attention heads to 4 in the NGSIM dataset leads to an enhancement in prediction accuracy. This improvement can be attributed to the dataset's challenging nature, wherein each attention head encapsulates a set of weights that captures specific aspects of the influence exerted by surrounding vehicles' interactions, along with spatial-temporal information consistency, impacting the future motions of the target vehicle. In contrast, the HighD dataset exhibits optimal prediction accuracy with fewer attention heads (specifically, 2 attention heads). By the combination of attention vectors, the model excels at extracting higher-order interactions among agents, which subsequently translates into improved predictive performance.

2) The proposed model is compared to various baseline models, including deterministic (S-LSTM and CS-LSTM) and more advanced stochastic models (M-LSTM, S-GAN, P-LSTM, HMNet, PiP, and MHA-LSTM), see subsection IV-D. The experiments have demonstrated that our proposed model (***Ours***) outperforms other stochastic predictors on both the NGSIM dataset and the HighD dataset, as indicated by the results presented in Table IV and Table V, respectively.

TABLE III

Optimising Number of Attention Heads for Improved Performance: $\mathcal{E}_{RMSE}$ in Meters (m) Across 5-Second Prediction Horizon for Different Number of Attention Heads on the HighD and NGSIM Datasets

Data	Metric	Time (s)	Heads (h) 2	4	8
NGSIM	$\mathcal{E}_{RMSE}(m)$	*1.0*	0.6270	**0.5385**	0.5929
		2.0	1.2996	**1.1328**	1.2237
		3.0	2.0548	**1.8074**	1.9351
		4.0	2.9639	**2.6339**	2.8116
		5.0	4.1008	**3.6877**	3.9137
HighD		*1.0*	**0.1918**	0.1901	0.1898
		2.0	**0.5504**	0.5493	0.5490
		3.0	**1.0973**	1.0991	1.1029
		4.0	**1.8072**	1.8087	1.8291
		5.0	**2.7271**	2.7694	2.7772

To this end, two metrics of $\mathcal{E}_{RMSE}$ and $\mathcal{E}_{NLL}$ are adopted for performance evaluation and to understand the model performance with comparability of in-between and aggregate level of comprehension. $\mathcal{E}_{RMSE}$ is a versatile and widely utilised validation metric in accurately measuring magnitudes of errors with high sensitivity to more significant errors, which is an optimal choice of data-driven approaches. Likewise, $\mathcal{E}_{NLL}$ is adopted to understand the fit and performance of the model with respect to the adopted probabilistic approach in comprehending how well the predicted probabilities match the actual distribution of the datasets.

The proposed model accurately predicts trajectories and recognises driving intents by incorporating social interaction and encoding spatio-temporal relationships between vehicles.

Our model excels in predicting long-term interactive trajectories, surpassing HMNet by effectively capturing the corresponding intention patterns among multiple interactive agents. Among the considered predictors, MHA-LSTM achieves the highest prediction performance due to its capacity to capture long-term driving intention patterns through the augmented multi-attention-based social pooling layer. However, it is worth noting that MHA-LSTM's interpretability may be constrained, as it does not account for the spatial correlations among the targets. In terms of average $\mathcal{E}_{RMSE}$ and the recognition of driving intents, our algorithm demonstrates a slight superiority over MHA-LSTM. Moreover, our performance surpasses that of the PiP algorithm with an average improvement of 11.8% and 6.2% on the NGSIM and HighD datasets, respectively.

The performance of our proposed model, as assessed by $\mathcal{E}_{NLL}$ values over a 5-second prediction horizon, exhibits a level of comparability, with a $\mathcal{E}_{NLL}$ of 5.81 at the 5-second prediction horizon.This can be validated that the developed model is conformable and performs well when integrated to the designed model framework evaluated with other approaches. This comparability of the $\mathcal{E}_{NLL}$ metric can be due to the hidden unobserved heterogeneity among the road users [78], [79] and their interaction which are not explicitly accounted for within the datasets adopted. Though, ours has a significant performance edge towards the $\mathcal{E}_{RMSE}$ for NGSIM and HighD datasets additional to better interpretability and performance. Notably, our model demonstrates superior performance in terms

TABLE IV

QUANTITATIVE EVALUATION OF THE LEARNING MODELS INCLUDING THE PROPOSED MODEL (OURS) ON THE NGSIM DATASET USING THE $\mathcal{E}_{RMSE}$ METRIC IN METERS (m) AND THE $\mathcal{E}_{NLL}$ METRIC IN ($nats$) OVER A 5-SECOND PREDICTION HORIZON. THE SMALLER THE VALUE, THE BETTER. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD** WHILE THE SECOND BEST RESULTS ARE MARKED WITH AN UNDERLINE

Data	Metric	Time (s)	M-LSTM [67]	HMNet [75]	S-LSTM [46]	CS-LSTM [6]	P-LSTM [29]	S-GAN [21]	PiP [28]	MHA-LSTM [34]	*Method 1*	*Method 2*	*Method 3*	*Method 4*	*Method 5*	*Method 6*	*Method 7*	***Ours***
NGSIM	$\mathcal{E}_{RMSE}(m)$	*1.0*	0.5931	**0.5003**	0.5780	0.5809	0.5960	0.5841	0.5568	0.5388	0.6207	0.6194	0.5965	0.6109	0.6293	0.5986	0.5403	<u>0.5385</u>
		2.0	1.2743	**1.1320**	1.2645	1.2678	1.3171	1.2765	1.1944	1.1387	1.2692	1.2521	1.2944	1.2677	1.2988	1.2197	1.1352	<u>1.1328</u>
		3.0	2.0984	1.8940	2.1078	2.1118	2.1892	2.1325	1.9519	1.8222	1.9930	1.9667	2.1452	1.9802	2.0409	1.9258	1.8130	**1.8074**
		4.0	3.1501	2.8574	3.1818	3.1796	3.3080	3.2256	2.9839	2.6644	2.9032	2.8568	3.2285	2.8562	2.9692	2.7848	2.6450	**2.6339**
		5.0	4.4607	4.0497	4.5275	4.5055	4.6757	4.5708	4.1036	3.7338	4.0702	3.9937	4.5766	3.9539	4.1608	3.8567	3.6990	**3.6877**
	$\mathcal{E}_{NLL}(nats)$	*1.0*	2.1488	–	**1.6508**	<u>1.6518</u>	1.6934	1.7232	1.9464	2.7246	2.3656	2.4075	2.1125	2.2872	0.2376	2.1890	3.6491	2.7907
		2.0	3.5770	–	<u>3.1432</u>	**3.1201**	3.1646	3.1899	3.3945	4.4892	3.6443	3.6789	3.5665	3.5569	2.3350	3.5562	5.1205	4.8672
		3.0	4.4009	–	3.9869	<u>3.9513</u>	3.9998	4.0199	4.2070	4.9595	4.3898	4.4214	4.4048	4.2999	**3.5842**	4.3060	5.4191	5.4553
		4.0	5.0250	–	<u>4.6077</u>	**4.5663**	4.6226	4.6475	4.8047	5.2313	4.9799	5.0118	5.0296	4.8976	4.4770	4.8836	5.8114	5.5739
		5.0	5.5562	–	<u>5.1282</u>	**5.0773**	5.1317	5.1675	5.3066	5.6321	5.5275	5.5530	5.5472	5.4430	5.1581	5.4234	6.0894	5.8127

Legend: *Method 1* (GRU-LSTM/Bi-LSTM) , *Method 2* (GRU/LSTM), *Method 3* (LSTM/LSTM), *Method 4* (GRU/Bi-LSTM), *Method 5* (LSTM/Bi-LSTM), *Method 6* (GRU/Bi-LSTM + *Fusion*), *Method 7* (GRU/Bi-LSTM+ MHA ($h = 1$).)

TABLE V

QUANTITATIVE EVALUATION OF THE LEARNING MODELS INCLUDING THE PROPOSED MODEL (OURS) ON THE HIGHD DATASET USING THE $\mathcal{E}_{RMSE}$ METRIC IN METERS (m) AND THE $\mathcal{E}_{NLL}$ METRIC IN ($nats$) OVER A 5-SECOND PREDICTION HORIZON. THE SMALLER THE VALUE, THE BETTER. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD** WHILE THE SECOND BEST RESULTS ARE MARKED WITH AN UNDERLINE

Data	Metric	Time (s)	M-LSTM [67]	PiP [28]	MHA-LSTM [34]	*Method 1*	*Method 2*	*Method 3*	*Method 4*	*Method 5*	*Method 6*	*Method 7*	***Ours***
HighD	$\mathcal{E}_{RMSE}(m)$	*1.0*	0.3288	**0.1867**	0.2247	0.329	0.2151	0.4421	0.1960	0.1844	0.4027	0.4104	<u>0.1918</u>
		2.0	0.8254	0.5557	0.6872	0.904	0.6051	1.03778	0.5629	**0.5459**	0.9142	0.9789	<u>0.5504</u>
		3.0	1.5700	1.1344	1.4144	1.747	1.1961	1.8727	1.1132	<u>1.0975</u>	1.6051	1.7643	**1.0973**
		4.0	2.5452	1.9215	2.3990	2.849	1.9790	2.9468	1.8526	<u>1.8404</u>	2.4519	2.6476	**1.8072**
		5.0	3.8022	2.9108	3.6222	4.200	2.9532	4.2117	<u>2.7837</u>	2.7838	3.4507	3.8243	**2.7271**
	$\mathcal{E}_{NLL}(nats)$	*1.0*	1.7129	<u>0.2631</u>	0.5677	1.007	0.3893	1.9900	0.3305	0.3970	0.8908	1.1718	**0.2184**
		2.0	3.3361	**2.3400**	2.7342	2.959	2.4007	3.4526	2.3667	2.8377	2.7005	3.2183	<u>2.2723</u>
		3.0	4.2880	**3.5844**	4.0133	4.140	3.6065	4.4206	3.5797	4.4693	4.2342	4.7256	<u>3.5205</u>
		4.0	5.0208	**4.4392**	4.9120	4.954	4.4716	5.1348	5.4546	5.5551	4.6640	6.1942	<u>4.4229</u>
		5.0	5.6909	**5.0977**	5.6683	5.577	5.1671	5.6801	5.6603	5.9278	5.4049	7.3087	<u>5.1515</u>

of $\mathcal{E}_{RMSE}$ while maintaining comparable $\mathcal{E}_{NLL}$ results when compared to various stochastic and deterministic methodologies.

It is important to note that the lower performance on the NGSIM data, compared to the results achieved with the HighD data, can be attributed to various factors, including flawed annotations, duplicated vehicles, noise, measurement errors, imbalanced manoeuvre classes [80], and perception issues [81].

3) Highway vehicles have non-holonomic motion systems constrained by their kinematic properties and the operating environment. Evaluating the precision of predictions for these systems relies on the $\mathcal{E}_{RMSE}$. However, $\mathcal{E}_{RMSE}$ has limitations as it averages all predictions, hiding significant deviations in lateral and longitudinal directions. Such deviations may not be accurately reflected in the overall $\mathcal{E}_{RMSE}$ value, leading to a skewed interpretation of the prediction error. Individually analysing a vehicle's lateral and longitudinal behaviour helps improve prediction precision. In the longitudinal direction (forward-backward), prediction errors can be relatively large, allowing for deviations of several meters due to fewer restrictions. However, the lateral (side-by-side) direction is more critical as stricter requirements are necessary to keep the vehicle within its lane. Lane width and vehicle dimensions influence predictability in different driving situations. A risk assessment framework considers an $\mathcal{E}_{RMSE}$ value close to $3m$ in the critical lateral direction. This indicates that the TV may barely maintain its lane and could potentially veer off-road within a few seconds, especially on highways with specific lane widths. Using the Euclidean distance to calculate the average $\mathcal{E}_{RMSE}$ may not accurately represent the observed variability, as anomalies and deviations can influence it in both lateral and longitudinal directions. Considering prediction error limits in both lateral and longitudinal directions is crucial for realistic vehicle behaviour predictions.This approach enhances our understanding of how the TV moves within its lane across various scenarios, allowing for a more accurate assessment of the prediction's physical realism.

Table VI and Table VII present our algorithm's prediction results, comparing the $\mathcal{E}_{RMSE}$ values computed for the longitudinal and lateral directions over a 5-second prediction horizon in NGSIM dataset and highD dataset, respectively.

Figures 4 and 5 show that the proposed approach demonstrates competitive performance against its counterparts when end-to-end trained on the NGSIM dataset and HighD dataset, respectively.

In Figure 3, we present a visualisation of prediction results in moderate traffic conditions using the NGSIM I-80 and US-101 datasets. Our model utilises a 3-second history of observed trajectories to predict future trajectories over a 5-second horizon.

B. Ablation Experiments

We conducted ablation studies to select network components and incorporate multiple encoder-decoder configurations in

TABLE VI

QUANTITATIVE EVALUATION OF THE LEARNING MODELS INCLUDING THE PROPOSED MODEL (OURS) ON THE NGSIM DATASET USING THE $\mathcal{E}_{RMSE}$ METRIC IN METERS (m) COMPUTED ALONG THE LONGITUDINALAND TRANSVERSE DIRECTIONS OVER A 5-SECOND PREDICTION HORIZON. THE SMALLER THE VALUE, THE BETTER. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD** WHILE THE SECOND BEST RESULTS ARE MARKED WITH AN UNDERLINE

Metric	Data	Time (s)	M-LSTM [67]	HMNet [75]	S-LSTM [46]	CS-LSTM [6]	P-LSTM [29]	S-GAN [21]	PiP [28]	MHA-LSTM [34]	*Method 1*	*Method 2*	*Method 3*	*Method 4*	*Method 5*	*Method 6*	*Method 7*	***Ours***
NGSIM	$\mathcal{E}_{RMSE}(m)$: *Longitudinal*	*1.0*	0.5751	**0.4848**	0.5613	0.5645	0.5796	0.5720	0.5251	0.5228	0.6033	0.5930	0.5783	0.6039	0.6112	0.5756	0.5241	0.5186
		2.0	1.2489	1.1108	1.2418	1.2452	1.2950	1.2662	1.1403	1.1186	1.2466	1.2288	1.2693	1.2478	1.2752	1.1990	1.1153	**1.0955**
		3.0	2.0688	1.8708	2.0829	2.0870	2.1635	2.1271	1.8740	1.8019	1.9682	1.9409	2.1161	1.9592	2.0142	1.9046	1.7930	**1.7549**
		4.0	3.1192	2.8338	3.1567	3.1548	3.2761	3.2275	2.7986	2.6443	2.8779	2.8302	3.1982	2.8350	2.9415	2.7637	2.6254	**2.5777**
		5.0	4.4300	4.0269	4.5034	4.4816	4.6522	4.6011	3.9443	3.7126	4.0446	3.9665	4.5465	3.9319	4.1326	3.8347	3.6784	**3.6266**
	$\mathcal{E}_{RMSE}(m)$: *Lateral*	*1.0*	0.1451	0.1237	0.1377	0.1370	0.1388	0.1411	0.1366	0.1302	0.1457	0.1465	0.1459	0.1374	0.1497	0.1375	*0.1316*	**0.1287**
		2.0	0.2529	0.2181	0.2385	0.2380	0.2405	0.2456	0.2410	0.2128	0.2383	0.2407	0.2534	0.2236	0.2460	0.2239	0.2118	**0.2080**
		3.0	0.3516	0.2958	0.3232	0.3222	0.3255	0.3304	0.3372	0.2715	0.3134	0.3177	0.3518	0.2880	0.3290	0.2848	0.2687	**0.2670**
		4.0	0.4398	0.3658	0.3990	0.3964	0.4008	0.4049	0.4242	0.3269	0.3818	0.3887	0.4410	0.3474	0.4048	0.3422	**0.3208**	0.3227
		5.0	0.5223	0.4295	0.4671	0.4632	0.4683	0.4719	0.5064	0.3964	0.4553	0.4655	0.5245	0.4172	0.4833	0.4112	**0.3898**	0.3916

TABLE VII

QUANTITATIVE EVALUATION OF THE LEARNING MODELS INCLUDING THE PROPOSED MODEL (OURS) ON THE HIGHD DATASET USING THE $\mathcal{E}_{RMSE}$ METRIC IN METERS (m) COMPUTED ALONG THE LONGITUDINALAND TRANSVERSE DIRECTIONS OVER A 5-SECOND PREDICTION HORIZON. THE SMALLER THE VALUE, THE BETTER. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD** WHILE THE SECOND BEST RESULTS ARE MARKED WITH AN UNDERLINE

Data	Metric	Time (s)	M-LSTM [67]	PiP [28]	MHA-LSTM [34]	*Method 1*	*Method 2*	*Method 3*	*Method 4*	*Method 5*	*Method 6*	*Method 7*	***Ours***
HighD	$\mathcal{E}_{RMSE}(m)$: *Longitudinal*	*1.0*	0.2923	**0.1653**	0.2197	0.3652	0.2102	0.4113	0.1779	0.1791	0.3992	0.3993	0.1865
		2.0	0.7692	**0.4962**	0.6747	0.9640	0.5922	0.9861	0.5330	0.5322	0.9035	0.9606	0.5365
		3.0	1.5067	**1.0201**	1.3970	1.8101	1.1776	1.8116	1.0785	1.0778	1.5880	1.7416	1.0775
		4.0	2.4817	**1.7212**	2.3781	2.9049	1.9563	2.8834	1.8075	1.8164	2.4295	2.6200	1.7828
		5.0	3.7428	**2.5912**	3.5975	4.2682	2.9259	4.1493	2.7511	2.7556	3.4227	3.7934	2.6985
	$\mathcal{E}_{RMSE}(m)$: *Lateral*	*1.0*	0.1506	0.0473	0.0472	0.0687	0.0457	0.1620	0.0486	**0.0438**	0.0527	0.0948	0.0447
		2.0	0.2994	0.1455	0.1305	0.1830	0.1239	0.3234	0.1261	**0.1215**	0.1392	0.1883	0.1225
		3.0	0.4415	0.2733	0.2217	0.3066	0.2095	0.4745	0.2105	**0.2068**	0.2342	0.2823	0.2073
		4.0	0.5650	0.4060	0.3159	0.4267	0.2988	0.6075	0.2970	0.2961	0.3310	0.3808	**0.2959**
		5.0	0.6697	0.5250	0.4218	0.5423	0.4003	0.7222	0.3949	0.3948	0.4381	0.4853	**0.3942**

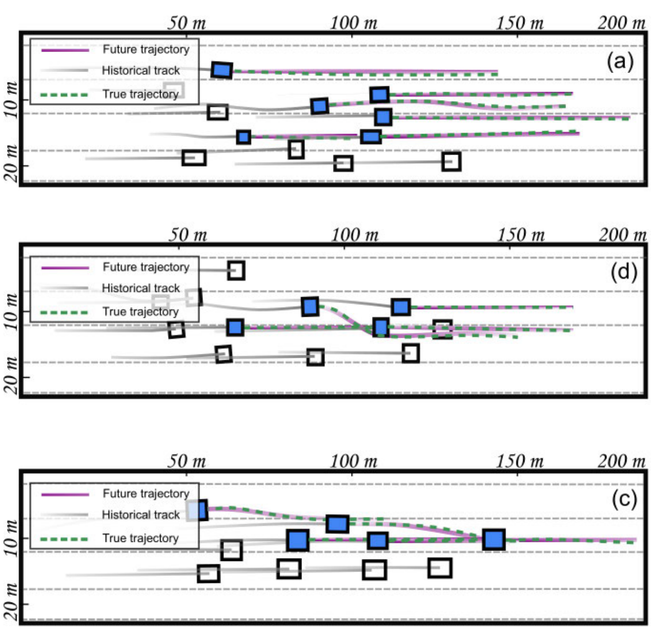

Fig. 3. **Diverse Trajectory Predictions for Vehicles:** three depictions (a, b, c) are given with lane change and lane-keeping manoeuvres. The ground truth (dashed green) and predicted trajectories (purple gradient) are visualised by sets of locations within the highway.

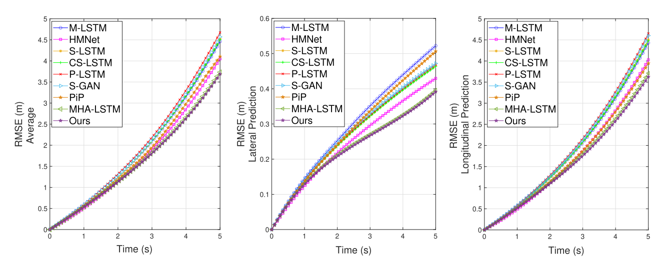

Fig. 4. The $\mathcal{E}_{RMSE}$ curves in meters *(m)* of end-to-end trained models on the NGSIM data: the average $\mathcal{E}_{RMSE}$ (left), lateral $\mathcal{E}_{RMSE}$ (middle), and longitudinal $\mathcal{E}_{RMSE}$ (right). The curve spans from the baseline to the future time.

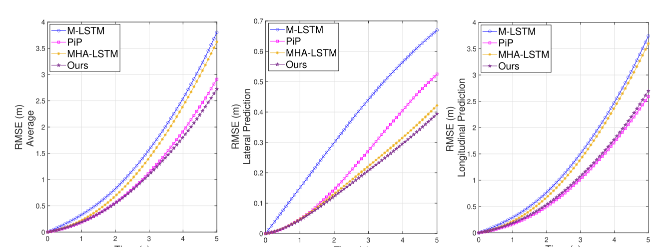

Fig. 5. The $\mathcal{E}_{RMSE}$ curves in meters *(m)* of end-to-end trained models on the HighD data: the average $\mathcal{E}_{RMSE}$ (left), lateral $\mathcal{E}_{RMSE}$ (middle), and longitudinal $\mathcal{E}_{RMSE}$ (right). The curve spans from the baseline to the future time.

designing an optimal learning algorithm. The studies involved comprehensive end-to-end training and testing through six experiments, each utilising specific datasets as presented below.

• ***Experiment 1.*** examines the encoding efficiency of two encoder heads (GRU and LSTM) with a Bi-LSTM decoder. The network utilises two encoder heads to capture both the dynamics of the TV and the past trajectories of SVs. This setup accurately depicts the positions of SVs relative to the TV. However, it may need help in learning motion features effectively.

- ***Experiment 2.*** unifies the encoder heads and utilises an LSTM decoder. The network uses the GRU to jointly encode the dynamics of TVs and historical motions of SVs. Both encoders share weights and parameters for both TVs and SVs. This setup enables compelling joint predictions by combining the inputs of multiple vehicles within a small sample, allowing for adaptability to an arbitrary number of vehicles during the prediction process [13].
- ***Experiment 3.*** compares two decoder heads, the GRU-LSTM network and the GRU-Bi-LSTM network. Both networks utilise the GRU to encode the dynamics and past trajectories of TV and SVs. The GRU-LSTM encoder-decoder and the GRU-Bi-LSTM encoder-decoder demonstrate superior results in terms of $\mathcal{E}_{RMSE}$. With the Bi-LSTM decoder, the network slightly outperforms the LSTM-based decoder. The Bi-LSTM's capability to process sequential data in both directions enables it to effectively capture context from both past and future information.
- ***Experiment 4.*** evaluates the efficiency of an LSTM encoder-decoder for predicting TV motions. *Method 3* utilises the LSTM encoder-decoder but performs poorly compared to other methods in the ablation study. This could be attributed to the LSTM's long-term memory capacity limitations, resulting in a subpar model. Furthermore, while the LSTM excels with dense input data, it may struggle with sparse sequential data, leading to lower prediction performance and limited generalisability.
- ***Experiment 5.*** evaluates the decoding efficiency of a GRU encoder and Bi-LSTM-MHA decoders with $h = 1$. The prediction performance is lower compared to the proposed approach that employs multiple heads. The decline can be attributed to the limitations in capturing complex dependencies with only one attention head and the computational cost of processing all data with a single attention mechanism.
- ***Experiment 6.*** evaluates the impact of the target fusion module on the prediction performance. The model consistently outperforms other models across the two considered datasets, indicating superior performance. These results highlight the effectiveness of incorporating target fusion in trajectory prediction, as it significantly improves the accuracy of multi-agent forecasting.

The experiments reveal that the most successful network configurations for trajectory prediction involve combining GRU or LSTM encoders with Bi-LSTM decoders. Additionally, using a GRU encoder with a Bi-LSTM decoder along with a MHA mechanism and the target fusion module, as proposed in our approach, proves to be effective. Among the evaluated datasets, the GRU-Bi-LSTM encoder-decoder achieves the highest performance. While the LSTM encoder-decoder and GRU-Bi-LSTM-MHA encoder-decoder (with a single head, $h = 1$) yield slightly lower results compared to the proposed approach.

C. Generalisability Test

Generalising to new inputs is a significant challenge for learning models in real-world applications. It requires achieving a balance between the best fit and avoiding overfitting. Generalisability refers to a model's ability to maintain performance on unseen or shifted domains [16]. Understanding how a model responds to different inputs helps address this issue [82]. Many studies assume an *identical* and *independent distribution* of training and test data. However, this assumption may not hold in real-world scenarios, where driving locations and traffic conditions can vary significantly. While increasing the size and diversity of the dataset can partially alleviate this issue, challenges such as data imbalance [83], [84] and long-tail cases with sparse training data can still result in inaccuracies and unreliable predictions.

To assess the generalisability of our approach, we propose utilising cross-data validation techniques. Initially, the model is trained and tested on data from the same highway traffic scenarios, establishing a benchmark or reference performance. The disparity in prediction accuracy, measured using an appropriate method [85], in comparison to the benchmark performance, indicates the model's generalisability and the extent of performance degradation. This procedure simulates an out-of-distribution scenario where the test data and scenarios differ partially or completely from the training data.

The internal architecture of the prediction model plays a crucial role in determining its generalisability, which is a key focus of this paper. In this subsection, we explore the influence of the model's internal architecture on its generalisability performance and analyse the observed degradation in performance in both lateral and longitudinal directions compared to the benchmark prediction. We have conducted the generalisability tests branched out into: (1) end-to-end training on HighD data and testing on NGSIM data; and (2) end-to-end training on NGSIM data and testing on highD data. The results are presented in the Table VIII and Table IX.

1) **End-to-End Training on HighD Data and Testing on NGSIM Data**

- Generalisability experiments show that *Methods* 2, 4, and 5 demonstrate good performance (Table VIII).
- *Methods* 4 and 5 exhibit comparable performance, consistently ranking among the top five most generalisable methods in the hierarchy of generalisability assessments. This is likely attributable to their reliance on a simple encoder-decoder configuration.
- Our method exhibits high generalisability proficiency, comparable to that of *Methods* 2 and 4. It should be pointed out that the methods previously outlined exhibit lower prediction proficiency. Our method achieves a well-balanced prediction and generalisability performance, providing an equitable mechanism which guarantees robust and long-term predictive and generalisation capabilities.
- Our method outperforms other competing social-interaction aware methods (PiP, HMNet, and MHA-LSTM) by integrating a GRU encoder and a hierarchical decoder comprising Bi-LSTM and MHA-based transformer. It shows particular proficiency in short- and medium-term prediction. During this timeframe, our method achieves an average $\mathcal{E}_{RMSE}$ of $1.57m$ at the 2-second prediction horizon, a lateral $\mathcal{E}_{RMSE}$ of $0.29m$, and a deviation of 30% from the benchmark performance. At the 5-second prediction horizon, it demonstrates an average $\mathcal{E}_{RMSE}$ of $5.12m$, a lateral $\mathcal{E}_{RMSE}$

TABLE VIII

QUANTITATIVE EVALUATION OF GENERALISABILITY FOR TRAJECTORY PREDICTION ALGORITHMS, INCLUDING OUR PROPOSED APPROACH (***Ours***). THE ALGORITHMS ARE TRAINED ON THE HIGHD DATASET AND TESTED ON THE NGSIM DATASET. GENERALISABILITY PERFORMANCE IS ASSESSED USING THE $\mathcal{E}_{RMSE}$ METRIC (IN METERS *(m)*) OVER A 5-SECOND PREDICTION HORIZON. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD** WHILE THE SECOND BEST RESULTS ARE MARKED WITH AN UNDERLINE

Metric	Dataset	Time (s)	M-LSTM [67]	PiP [21]	HMNet [75]	MHA-LSTM [34]	*Method 2*	*Method 3*	*Method 4*	*Method 5*	*Method 6*	***Ours***
$\mathcal{E}_{RMSE}(m)$	HighD* NGSIM	*1.0*	0.7936	0.7813	0.7886	0.7886	**0.7180**	0.8164	0.7446	0.7290	0.8096	0.7228
		2.0	1.6659	1.7160	1.7071	1.7070	**1.5292**	1.7442	1.6167	1.5644	1.7672	1.5754
		3.0	2.6638	2.8165	2.7720	2.7719	**2.4247**	2.8023	2.5960	2.5200	2.7833	2.5495
		4.0	3.8772	4.1426	4.0587	4.0585	**3.5032**	4.0717	3.7487	3.6847	3.9364	3.7159
		5.0	5.3457	5.7007	5.5776	5.5775	**4.8239**	5.5908	5.1006	5.1394	5.3938	5.1210
$\mathcal{E}_{RMSE}(m)$: *Longitudinal*	HighD* NGSIM	*1.0*	0.7710	0.7504	0.7444	0.7700	0.7247	**0.6996**	0.7950	0.7105	0.7916	0.7031
		2.0	1.6351	1.6610	1.6197	1.6803	**1.5037**	1.7161	1.5893	1.5385	1.7395	1.5480
		3.0	2.6282	2.7485	2.6344	2.7438	**2.3976**	2.7697	2.5676	2.4931	2.7532	2.5209
		4.0	3.8399	4.0695	3.8727	4.0312	**3.4770**	4.0373	3.7211	3.6592	3.9072	3.6882
		5.0	5.3084	5.6254	5.3567	5.5504	**4.7980**	5.5559	5.0734	5.1144	5.3650	5.0936
$\mathcal{E}_{RMSE}(m)$: *Lateral*	HighD* NGSIM	*1.0*	0.1881	0.2174	0.1671	0.1703	**0.1614**	0.1857	0.1713	0.1633	0.1693	0.1678
		2.0	0.3188	0.4310	0.2923	0.3004	**0.2780**	0.3118	0.2963	0.2835	0.3102	0.2927
		3.0	0.4343	0.6150	0.3856	0.3935	**0.3612**	0.4264	0.3829	0.3669	0.4080	0.3809
		4.0	0.5365	0.7746	0.4640	0.4703	**0.4273**	0.5280	0.4536	0.4322	0.4778	0.4532
		5.0	0.6310	0.9237	0.5432	0.5495	**0.4998**	0.6243	0.5257	0.5056	0.5561	0.5292

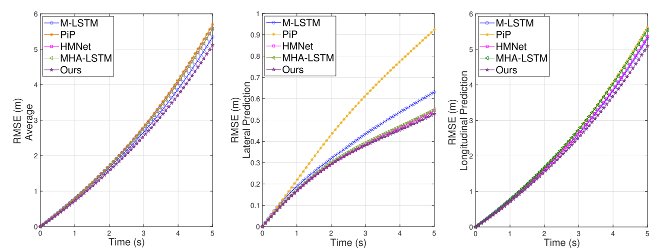

Fig. 6. The curve depicts predicted $\mathcal{E}_{RMSE}$ values *(m)* as an indicator of generalisability performance for different models. All models were end-to-end trained on the HighD and tested on the NGSIM. The $\mathcal{E}_{RMSE}$ is calculated for three measures: average $\mathcal{E}_{RMSE}$ (left), lateral $\mathcal{E}_{RMSE}$ (middle), and longitudinal $\mathcal{E}_{RMSE}$ (right). The curve spans from the baseline to the future time.

of $0.52m$, and a performance drop of 26% relative to the benchmark performance.

- *Method* 6 exhibits a slight drop in prediction performance. The observed decrease in performance when generalising across different datasets may be influenced by the sensitivity of the CNN to noise [86]. This assumption is based on the understanding that CNN-based models can be susceptible to noise, which may impact the network's ability to generalise effectively, particularly when incorporating target fusion information.
- In this investigation, we noted that the transformer encoder-decoder architecture employing multi-headed attention-based demonstrates a commendable capacity for adaptation. This was particularly evident in its ability to encompass various crucial aspects of predicted trajectories. This adaptability was consistently demonstrated even when the model was tested with different datasets. Specifically, our method yielded satisfactory results when trained with fewer attention heads (i.e., some heads are universally important and others appear to contribute less to the model [77]) when applied to a dissimilar dataset that necessitated a more substantial attentional focus for optimal performance. This demonstrates our method's adaptation and robustness across various naturalistic driving datasets.

2) **End-to-End Training on NGSIM Data and Testing on HighD Data**

As for generalisability, we evaluate various trained approaches and combinations (some combinations are omitted here for brevity). This experiment evaluates an end-to-end model on the NGSIM (US-101 and I80) datasets using unseen traffic data from highD. This highD dataset exhibits dissimilar traffic behaviour due to cross-cultural differences in driving behaviours, road geometry, and traffic regulations. The results are presented in the table below (Table IX).

- Consistent with previous findings, the proposed model demonstrates enhanced performance when trained on highD data and tested on NGSIM data. However, the results obtained in this study highlight clear fallibility and insignificance across all models, including encoder-decoder combinations and the proposed approach, in predicting trajectories over medium- and long-term horizons. Notably, the failures in prediction primarily stem from longitudinal prediction. Our approach achieves an accuracy of $0.15m$ at 1-second and $0.43m$ at a 3-second prediction horizon for longitudinal prediction, which falls within acceptable limits for lateral prediction. When considering the averaged outcomes of both lateral and longitudinal predictions, it becomes apparent that the prediction results lack logical consistency and exhibit significant deviations from the ground truth. These impractical deviations, characterised by substantial distances, are inconsistent with the actual behaviour of vehicles in the medium- and long-term predictions.
- The proposed approach demonstrates good generalisability performance within a prediction horizon of less than 2-second of prediction horizon. It achieves an accuracy of $7.07m$ and $0.15m$ in the longitudinal and lateral directions, respectively, with an average error of $7.07m$. This outcome indicates that the prediction model effectively anticipates trajectories over this short-term horizon. However, as the prediction horizon expands, the models noticeably diverge, leading to exponential growth in prediction error. This is likely attributed to:

a) ***Behavioural Contrasts in Traffic Scenarios.*** The two traffic scenarios demonstrate significant behavioural disparities. For instance, in the less dynamic HighD data, there is a higher frequency of lane-keeping manoeuvres, indicating a lower

TABLE IX

QUANTITATIVE EVALUATION OF GENERALISABILITY FOR TRAJECTORY PREDICTION ALGORITHMS, INCLUDING OUR PROPOSED APPROACH (OURS). THE ALGORITHMS ARE TRAINED ON THE NGSIM DATASET AND TESTED ON THE HIGHD DATASET. GENERALISABILITY PERFORMANCE IS ASSESSED USING THE $\mathcal{E}_{RMSE}$ METRIC (IN METERS (m)) OVER A 5-SECOND PREDICTION HORIZON.THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD** WHILE THE SECOND BEST RESULTS ARE MARKED WITH AN UNDERLINE

Metric	Dataset	Time (s)	M-LSTM [67]	PiP [21]	MHA-LSTM [34]	*Method 2*	*Method 3*	*Method 4*	*method 5*	*method 6*	***Ours***
$\mathcal{E}_{RMSE}(m)$	NGSIM* HighD	*1.0*	7.7541	**6.4155**	7.2757	7.9383	7.7753	7.3851	6.9937	9.3225	7.0741
		2.0	15.7385	**12.5188**	14.3888	14.8925	15.3894	13.8069	14.4199	19.0125	14.5559
		3.0	23.9983	**18.7395**	22.0053	22.1778	22.9871	20.5657	21.8650	28.9971	22.3858
		4.0	32.4036	**25.0242**	30.0133	29.7564	30.6382	27.3509	29.4531	39.1479	30.6098
		5.0	40.6759	**31.0986**	38.1100	37.2753	38.0790	33.3412	37.7951	49.5590	39.0005
$\mathcal{E}_{RMSE}(m)$: *Longitudinal*	NGSIM* HighD	*1.0*	7.7526	**6.4123**	7.2739	7.9367	7.7740	7.3836	6.9924	9.3215	7.0724
		2.0	15.7359	**12.5128**	14.3858	14.8891	15.3864	13.8038	14.4174	19.0107	14.5527
		3.0	23.9947	**18.7317**	22.0010	22.1731	22.9826	20.5612	21.8615	28.9947	22.3814
		4.0	32.9947	**25.0151**	30.0080	29.7508	30.6325	27.3454	29.4486	39.1450	30.6047
		5.0	40.6706	**31.0887**	38.1038	37.7583	38.0722	33.3348	37.7899	49.5557	38.9950
$\mathcal{E}_{RMSE}(m)$: *Lateral*	NGSIM* HighD	*1.0*	0.1517	0.2027	0.1637	0.1598	0.1432	0.1483	0.1397	**0.1385**	0.1511
		2.0	0.2888	0.3861	0.2907	0.3145	0.3008	0.2915	0.2645	**0.2611**	0.3026
		3.0	0.4180	0.5410	0.4326	0.4565	0.4536	0.4280	0.3918	**0.3744**	0.4393
		4.0	0.5376	0.6738	0.5652	0.5791	0.5919	0.5475	0.5152	**0.4758**	0.5571
		5.0	0.6536	0.7849	0.6858	0.6838	0.7177	0.6485	0.6289	**0.5759**	0.6575

TABLE X

COMPUTATION TIME IN *seconds* (S): COMPUTATION EFFICIENCY OF THE PROPOSED APPROACH AND THAT OF COMPARISON APPROACHES. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD** WHILE THE SECOND BEST RESULTS ARE MARKED WITH AN UNDERLINE

Method	M-LSTM [67]	HMNet [75]	S-LSTM [46]	CS-LSTM [6]	P-LSTM [29]	S-GAN [21]	PiP [28]	MHA-LSTM [34]	***Ours***
Batch Size = 1 *Time(s) (±std)*	0.0080 (±0.0122)	**0.0021** (±0.0010)	0.0040 (±0.0017)	0.0039 (±0.0022)	0.0043 (±0.0021)	0.0041 (±0.0023)	0.0189 (±0.9812)	0.0143 (±0.0449)	0.0234 (±0.01123)
Batch Size = 64 *Time(s) (±std)*	0.0207 (±0.1022)	**0.0028** (±0.0050)	0.0050 (±0.0019)	0.0049 (±0.0053)	0.0234 (±0.0112)	0.0367 (±0.0076)	0.4659 (±0.1387)	0.0296 (±0.0640)	0.3188 (±0.0743)
Batch Size = 128 *Time(s) (±std)*	0.0282 (±0.1130)	**0.0029** (±0.0074)	0.0051 (±0.0019)	0.0051 (±0.0105)	0.0704 (±0.0235)	0.0702 (±0.0047)	0.7433 (±0.2042)	0.0385 (±0.0961)	0.5972 (±0.1060)

occurrence of lane changes. However, this characteristic is not necessarily observed in the NGSIM I-80 data [80].

b) ***Imbalanced learning.*** real-world datasets often exhibit an imbalance, where the minority class, which is of primary interest for detection or recognition, contains infrequent occurrences. To address this issue, two approaches can be utilised: sampling methods and ensemble learning. These techniques aim to alleviate the impact of class imbalance and enhance model performance on imbalanced datasets. They target undersampled instances, such as lane change manoeuvre classes, while selectively reducing the number of oversampled instances, such as lane keeping manoeuvre classes [84].

c) ***Other.*** the observed disparity can be attributed to other factors as well. One possible reason is the substantial difference in scale between the highD and NGSIM datasets, with highD being notably more extensive. Furthermore, noise and inaccuracies in the annotation process within the NGSIM dataset can lead to unrealistic vehicle behaviours.

D. Computation Time

Efficient computation plays a crucial role in multi-agent learning systems for automated driving. We assessed the processing capabilities of our approach with time-varying inputs through a comprehensive analysis of computation time. The summarised results can be found in Table X.

This table shows computation times for the algorithms with varying batch sizes (i.e., *batch_size=1, 64, 128*). For a *batch_size=128*, CS-LSTM, S-GAN, and P-LSTM are 14× faster ($0.005s$) than P-LSTM and S-GAN ($0.07s$) for trajectory prediction on the NGSIM test set. Our approach takes $0.59s$, while HMA-LSTM is 15× faster ($0.03s$) to generate final predictions. HMNet predicts in $0.002s$, while our approach is 10× slower. PiP method has the longest prediction time due to its resource-intensive planning and fusion modules for marginal improvements. With limited resources, our approach currently demands additional computational time compared to other methods. However, allocating more resources can enhance the model's performance when integrating it into a real-time AV system. Furthermore, our approach exhibits increased computation time when the batch size is increased. This, however, does not affect real-time computation. Consequently, when the *batch_size* = 1, our approach can effectively compete with other equally effective methods (e.g., MHA-LSTM [34]) in terms of computational efficiency. It can also be inferred that the computational time is a trade-off to the model efficiency for longer horizon predictability and batch sizes. There lies a hidden endogeneity in the run time, which can be affected by various factors like the used resource processor efficiency, code execution, benchmarking and other unobserved indicators which can affect the run time of the approach. This approach is still considered a pilot concept, and further testing on a real AV prototype is planned as future work.

VI. CONCLUSION

Accurate motion prediction is a vital aspect of AV operation as it allows the AV to understand and interpret its surroundings by combining multiple sensory observations and precise localisation. This helps the AV anticipate potential hazards in the road scene. The output of motion prediction is also crucial for subsequent processes such as decision-making and path planning. However, the efficiency of motion prediction can be challenging due to the many dependencies on various factors, including various uncertainties (e.g., data-dependent and task-dependent uncertainties), primarily when operating in complex and highly interactive traffic conditions. These factors can affect prediction performance and generalisation. AVs must infer the intentions of other vehicles, consider uncertainties, and make highly accurate predictions of their future trajectories based on learned social interactions. This research proposes a trajectory and intent prediction framework that integrates a temporal-spatial understanding of the surrounding environment, including social interactions among other vehicles, with a multi-modal prediction generation of potential future outcomes. These contributions significantly enhance prediction performance. The proposed framework implements a prediction refiner based on a multi-head attention mechanism, where each attention head is specialised in interpreting various features characterising the motion of target vehicles with a specific class of driving intent without requiring explicit labelling. The ablation experiment provides deeper insights into the crucial choice of the network's internal components, which can be essential for improving prediction and generalisation performance. Compared to existing approaches, the proposed framework improves trajectory prediction performance over a long-term prediction of $3 \sim 5$s in the future. The highest accuracy was achieved when trained on the highD dataset and tested on the NGSIM datasets, according to the generalisation experiment. In future research, we will use the current trajectory prediction results to re-examine high-level social interaction using graph theory to improve long-term trajectory prediction. It is important to note that the generalisation and transferability of prediction algorithms are critical for the widespread deployment of AVs on roads and, therefore, deserve further research efforts.

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